MIL-C-60174(MU)
AMENDMENT 3
12 July 1968
SUPERSEDING
AMENDMENT 2
23 April 1968

MILITARY SPECIFICATION

CONNECTOR, ELECTRICAL, RECEPTACLE: 10526125

This amendment forms a part of Military Specification MIL-C-60174(MU), dated 19 March 1965.

Paragraph 2.1: Delete the following specifications:

"MIL-C-5015 Connectors, Electric, AN Type
MIL-C-8384 Connectors, Plug and Receptacle, Electrical: General Specification for"

and substitute

"MIL-C-21097 Connectors, Electrical, Printed Wiring Board, General Purpose, General Specification for"

Paragraph 3.1.1: Delete and substitute:

"Insertion force.- The insertion force required to insert an .066 steel gage after five (5) insertions shall not exceed 48 pounds and in addition, the insertion force required to insert an .066 steel gage after ten (10) insertions of an .071 circuit board shall not exceed 40 pounds."

Paragraph 3.1.2, third sentence: Delete "0.005 inch" and substitute "0.008 inch".

Paragraph 3.2.1: Delete "0.061 to 0.063" and substitute "0.061 to 0.062"; Delete "4 to 8 ounces" and substitute "3 to 8 ounces"; Delete "-of the board."

Paragraph 3.4.3.1: Delete "3.5.2" and substitute "3.5".

* Paragraph 3.4.4: Delete and substitute:

"3.4.4 Salt spray.- Mated connectors shall be capable of withstanding Test Condition B of MIL-STD-202, Method 101. Subsequent to such exposure, the connector shall not have been damaged or adversely affected and, after being washed and thoroughly dried in accordance with MIL-C-21097, shall meet the requirements of 3.5 except that the contact resistance requirements in 3.5.3 shall cause less than 0.040 volts drop across each contact with 3.0 amperes dc flowing through the contact from an open circuit voltage source of 1.5 volts dc following insertion and withdrawal of a 0.054 inch to 0.056 inch thick goldplated board.

FSC 1220

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* Paragraph 4.1: Delete and substitute new paragraph 4.1:

"4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are necessary to assure supplies and services conform to prescribed requirements."

* Following paragraph 4.1.1, Add:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) connectors manufactured in the same manner using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.3: Add the following sentence: "No lots shall be accepted until control samples have been satisfactorily tested."

Paragraph 4.2.4: Delete and substitute:

"4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on at least two (2) of every 1000 connectors produced except 4.5.3.1 should be inspected on at least 2 out of each day's production.

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TABLE I

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
101. Contact resistance	3.2.1, 3.5.3	4.5.2.3
102. Insertion and separation forces (cycled)	3.1.1	4.5.3.2
103. Insertion and separation forces (uncycled)	3.2.1	4.5.3.1
104. Contact float	3.1.2	4.5.3.3

Paragraphs 4.3.2.1, 4.3.2.2, and Table III: Delete.

* Under paragraph 4.3.2 add:

"Connectors shall be selected at random by a Government representative as control samples from each three (3) months production or portion thereof as follows:

3 units per first 100 produced
5 units per 200
7 units per 300
9 units per 400
10 units per 500
12 units per 700
15 units per 1000
and 2 units for each 100 items produced in excess of 1000."

The control samples shall have met the requirements and tests of Tables I and II and shall then meet the requirements and tests of Table IV except that Characteristic 308 need only be performed on the first three (3) samples.

Table IV: Add the following:

"313. Vibration	3.4.1	4.5.1.1
314. Shock	3.4.2	4.5.1.2"

Paragraph 4.4.1.1:

Item 3: Delete "0.062 ± 0.001" and substitute "$0.062 \begin{smallmatrix}+.000\\-.001\end{smallmatrix}$"

Item 8: Delete "Maximum thickness of finished board shall not exceed 0.071 inch" and substitute "Thickness of finished board shall be $0.071 \begin{smallmatrix}+.000\\-.001\end{smallmatrix}$ inch".

Add "Item 10 Steel gage.- Outline as shown in Figure 1".

* Paragraph 4.5.1.1(b): Delete "four hours" and substitute "six hours".

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* Paragraph 4.5.1.3: Delete "4.5.2.2" and substitute "4.5.2".

* Paragraph 4.5.1.4.1: Delete and substitute:

"Measurements after exposure.- Within five minutes after this test, the exterior surfaces of the connectors shall be washed with tap water and dried in accordance with MIL-C-21097. At the end of the drying cycle, the connectors shall be subjected to the contact resistance tests defined in 4.5.2.3 of this specification except that the contact resistance shall cause 0.040 volt drop across the contact maximum with 3 amperes flowing through the contacts as required by 3.4.4 and 3.5.3."

Paragraph 4.5.1.9: Delete "one of the first 5 units" and substitute "two of the first 5 units".

Paragraph 4.5.2.3: Delete "Figure 2 of MIL-C-8384" and substitute "Figure 1a of MIL-C-21097".

Paragraph 4.5.2.3.1: Add the following: "When the contact resistance is being measured subsequent to an environmental test, the connector shall not be mated 500 times with a PC board and only one measurement of contact resistance shall be taken.

Paragraph 4.5.3.1: Delete and substitute:

"4.5.3.1 Insertion and separation forces (uncycled).- After 5 insertions of an .066 gage, the insertion force of the .066 gage shall not exceed 48 pounds and the withdrawal force of each contact pair of an .062 board shall be a minimum of 4 ounces."

Paragraph 4.5.3.2: Delete and substitute:

"4.5.3.2 Insertion and separation forces (cycled).- Insert an .066 steel gage for 500 cycles, after which an .062 printed circuit board, of such a width as to engage only one pair of contacts shall be inserted. The withdrawal force of any contact pair shall be 4 ounces minimum. Insert an .071 printed circuit board for ten cycles, after which the insertion force of an .066 steel gage shall not exceed 40 pounds. Insert an .062 printed circuit board of such a width as to engage only one pair of contacts. The withdrawal force of any contact pair shall be 3 ounces minimum."

* Paragraph 4.5.3.3: Delete last sentence and substitute "A 10 pound pulling force, directly on the contact terminal, shall not distort, bend or warp the contact surface. At least 4 contacts shall be selected at random for this test."

Add: Figure 1.

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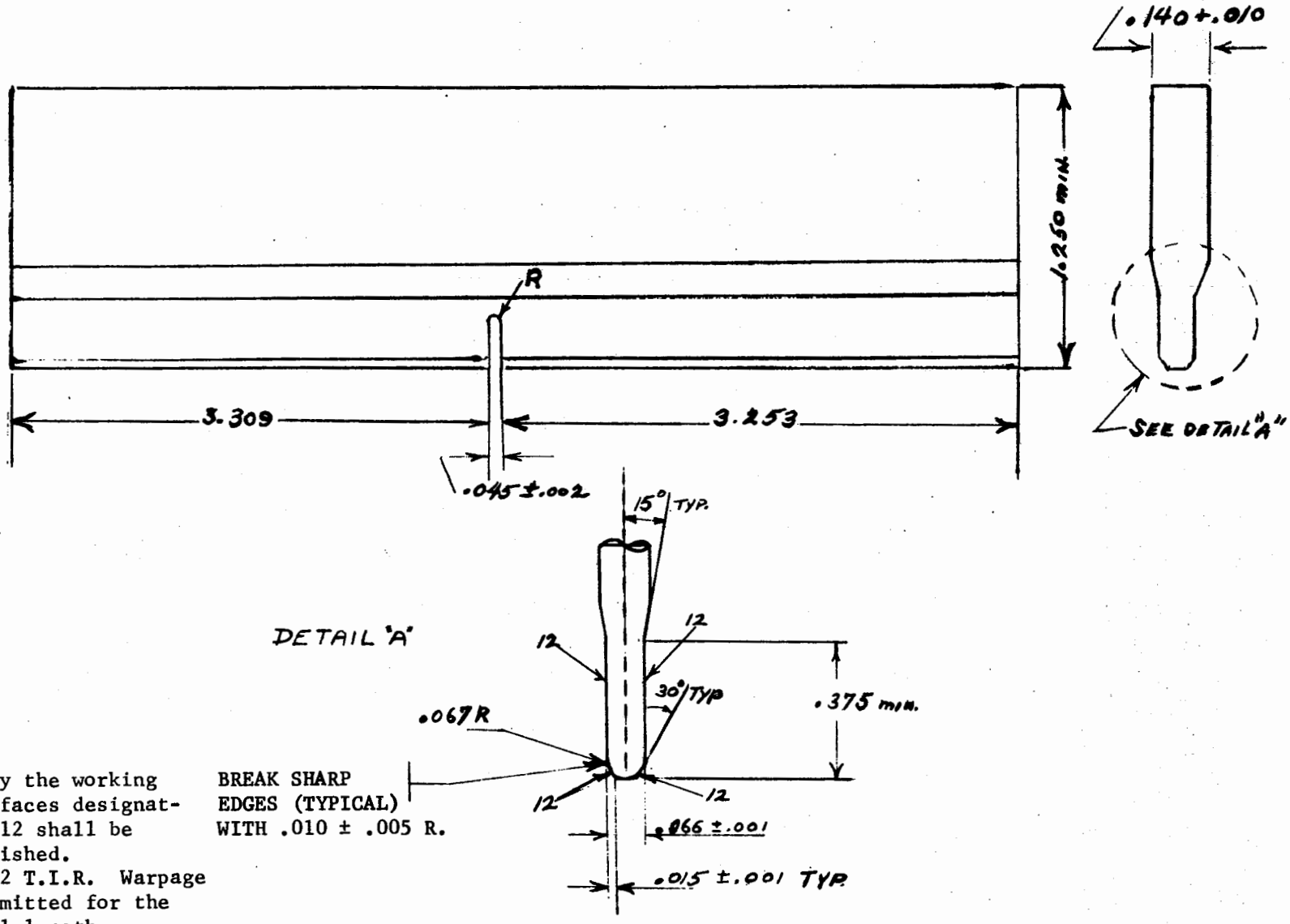
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NOTES:

1. Only the working surfaces designated 12 shall be finished.
2. .002 T.I.R. Warpage permitted for the full length.
3. Tolerance: ±.005 unless otherwise shown.
4. Material: Hardened tool steel Rockwell "C" 50-55

BREAK SHARP EDGES (TYPICAL) WITH .010 ± .005 R.

DETAIL A

FIGURE 1 MECHANICAL GAGE

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